Jeroen Buil

(BIOMEDICAL) AI ENGINEER

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Nationality: Netherlands | B - Work permit
Portfolio: www.jeroenbuil.github.io

11+ years of AI R&D experience in the biotech, medical and finance domains.
Driven to leave a positive impact with my work and at my work.
In my free time you will likely find me on the water or tinkering with electronics.

EXPERIENCE

Jan. 2025 – Present

Freelancer + Volunteer | *Switzerland*

- Freelance work to help SME and startups navigate regulations, requirements and applicable tech.
- Volunteering both for Bee & Biodiversity conservation and to help out local communities in need.

Mar. 2023 – Nov. 2024

Lead ML Engineer | *Wageningen Research* | *The Netherlands*

- Led AI project teams; handled stakeholder communication and day to day operation (biotech field).
- Hands-on coding on data platforms (ETL pipelines, (gen)AI models, dashboards) for decision making.
- Technical documentation. Ensuring coding+data standards (e.g. F.A.I.R) and mitigating discrepancies.

Keywords: project management, (gen)AI, Python, SQL, Javascript, Tableau, API, CI/CD, Cloud, F.A.I.R.

May 2021 – Feb. 2023

Senior Data Scientist | *ABN AMRO Bank* | *The Netherlands*

- Project lead applying big data analysis for financial crime and fraud detection.
- Product Owner + Developer of AI tooling for integrating core-data streams into data driven decisions.
- Coordinating mitigating strategies with management, compliance & tech teams. Mentored interns.
  Keywords: data science, SQL, Python, big data, time series analysis, ML, Databricks, PowerBI, Azure

Aug. 2018 – May 2021

Biomedical R&D Engineer (Algorithms) | *imec* | *The Netherlands*

- Medical (wearable) device validation: hands-on data analysis of sensor output for disease diagnosis.
- Algorithm, ML model and dashboard development for biosignal processing and interpretation.
- Technical documentation. Literature Research. Setting up clinical trials. Mentoring new colleagues.

Keywords: medical devices, wearable sensors, MATLAB, Python, AI, GDPR, ISO 13485, IEC 62304

Oct. 2017 – May 2018

Project Lead & Integration Engineer | *German Primate Center (DPZ)* | *Germany*

- Led a team developing an ultra-light collar, lowering costs/weight 10-30x, easing animal stress.
- Published PCB/CAD design, Arduino firmware, BLE link, and Android app for remote access.

Dec. 2013 – Oct. 2017

Neuroscience Researcher | *German Primate Center (DPZ)* | *Germany*

- Neuroprosthetics: Integrating robotics, neural decoders/stimulators with thin-film electrodes.
- Supervision of 5 MSc thesis projects. Elected PhD representative at the graduate school.

Keywords: biomedical engineering, brain-machine interfaces, robotics, ML, time series, motion tracking

EDUCATION

Oct. 2017 - PhD Neural Engineering (Magna Cum Laude) | University of Göttingen, DE

- Thesis: Peripheral Nervous System Control for Neuroprostheses

Jul. 2013 - MSc Biomedical Engineering (Cum Laude) | ETH Zürich, CH

- Thesis: Wearable device for assisting stroke patient rehabilitation

Jul. 2011 - BSc Biomedical Engineering (Cum Laude) | Technical University Eindhoven, NL

- Honours Programme graduate
- Minor Neuroscience - University of Nothingham – UK

SCHOLARSHIPS & HONOURS

Sep. 2011 - VSBfonds Scholarship

- One the most coveted scholarships available in The Netherlands for studying abroad.

Jul. 2011 - Recommendation letter of the Rector Magnificus – Technical University Eindhoven

- Rewarded for finishing my Bachelor degree with the highest average grade of my year.

Jul. 2011 - Honours Diploma

- Rewarded after successfully completing the Honours Programme.

PERSONAL SKILLS

Leadership

- My experience in different fields has given me profound understanding of the work my team members are doing, and what needs to be done to successfully finish a project. We generate results and still like each other afterwards. I am proud that colleagues call me for advice.

Analytical

- Years of experience in the analysis of a wide variety of data and signal types has taught me to recognise the strengths and weaknesses in strategies and designs.

Creative

- I get a lot of enjoyment out of finding new solutions or ways to improve current ones. This is reflected in the professional projects I worked on, as well as my choice of hobbies.

Computer skills

- Coding languages: Python, MATLAB, R, SQL, Java, JavaScript (React) and C++
- Data Visualization (Matplotlib, Seaborn, Plotly, PowerBI), Machine Learning (scikit-learn, Pytorch), Big Data (Databricks), Cloud Services (Azure, AWS) and ETL Pipelines (Airflow)
- Data modelling tools (SQL Server Management Studio, Azure Data Studio, MS Visio)
- Version control (Git), integration (CI/CD, API), containerisation (Docker)
- Neural data processing (MNE, EEGLAB)
- CAD Design (Solidworks, Fusion360) and PCB design (KiCAD)

LANGUAGES & INTERESTS

- Dutch: C2
- English: C1
- German: B1
- French: A2

- Photo/videography
- Wing foiling + Snowboarding
- Sim Racing
- Active member in the local MakerSpace (= spending time building stuff I could just buy in stores)